INCH-POUND

MIL-PRF-1/1389G
16 June 2020
SUPERSEDING
MIL-PRF-1/1389F(NAVY)
22 July 1999

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY
TYPE 10ANP7

Inactive for new design
after 21 June 2011

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: 10 inch, magnetic deflection, magnetic focus cathode ray tube with aluminized screen.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

Parameter: Unit:	Ef V	Ec1 V dc	Ec2 V dc	Eb V dc	Ehk V dc	Rg1 MegΩ	Alt ft
Maximum: Minimum:	6.9 5.7	0 -200	450 ---	11,000 7,000	+125 -125	1.5 ---	20,000 ---
Test condition:	6.3	Adj 1/	250	9,000	---	---	---

See footnotes at end of table I.

GENERAL:

Qualification - Required.

This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

AMSC N/A

FSC 5960

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TABLE I. Testing and inspection.

Inspection	MIL-STD-1311 Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
Qualification inspection							
Cathode illumination	5216	---		---	---	---	---
Direct-interelectrode capacitance	1331	---	k to all g1 to all	Ck Cg1	--- ---	5.0 6.3	pF pF
Electrode current (grid No. 2)	5201	---	Ec1 = 0	Ic2	---	5.0	μA dc
Pressure (implosion)	1141	---		---	---	---	---
Barometric pressure, reduced	1002	6/		---	---	---	---
Vibration	5111	---		Width	---	2.0	mm
Conformance inspection, part 1							
Voltage breakdown	5201	---		---	---	---	---
Voltage breakdown (magnetic types)	5201	---		---	---	---	---
Gas ratio	5206	5/		Gr	---	0.25	---
Neck straightness	5101	2/		---	---	---	---
Screen and faceplate blemishes	5106	---		---	---	---	---
Modulation	5223	1/	Ib = 400 μA dc	ΔEc1	---	40	V dc
Spot position (magnetic deflection)	5231	---		Radius	---	12.5	mm
Grid cutoff voltage	5241	---		Ec1	20	45	V dc
Focus coil current	5246	---	Ib = 100 μA dc; D = 2.5 inch using JEDEC No. 127 focus coil, or equivalent	---	40	60	mA dc
Aperture alignment	---	4/		---	---	8	mm
Grid No. 1 leakage current	5251	---		---	---	---	---

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	MIL-STD-1311 Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 2							
Heater current	1301	---		If	270	330	mA
Neck and bulb (magnetic types)	5101	---		---	---	---	---
Side terminal and base alignment	5101	---	Halfway between 1 and 9	---	-10	+10	Degrees
Face tilt	5101	---		---	---	---	---
Stray light emission (conventional types)	5216	---	Eb = 11,000 V dc; Ec2 = 450 V dc	---	---	---	---
Screens	5221	3/		---	---	---	---
Line width A (magnetic deflection)	5226	---	Ib = 100 µA dc	Width	---	0.33	mm
Line width C (magnetic deflection)	5226	---	Ib = 100 µA dc	Width	---	0.50	mm
Heater-cathode leakage current	5251	---		---	---	---	---
Grid No. 2 leakage current	5251	---		---	---	---	---
Permanence of marking	1105	---		---	---	---	---
Conformance inspection, part 3							
Life test	---	---	Group C; Eb = 11,000 V dc; Ec2 = 450 V dc; Ib = 60 µA dc	t	500	---	hrs
Life-test end points:	---						
Line width A	5226	---	Ib = 60 µA	Width	---	0.33	mm
Modulation	5223	1/	Ib = 320 µA	ΔEc1	---	40	V dc

See footnotes at top of next page.

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TABLE I. Testing and inspection - Continued.

1/ ΔEc1 shall be not greater than the value of grid cutoff voltage (see method 5241).

2/ Cylinder shall be 4 inches (101.60 mm) long and .910 inch (23.11 mm) maximum inside diameter.

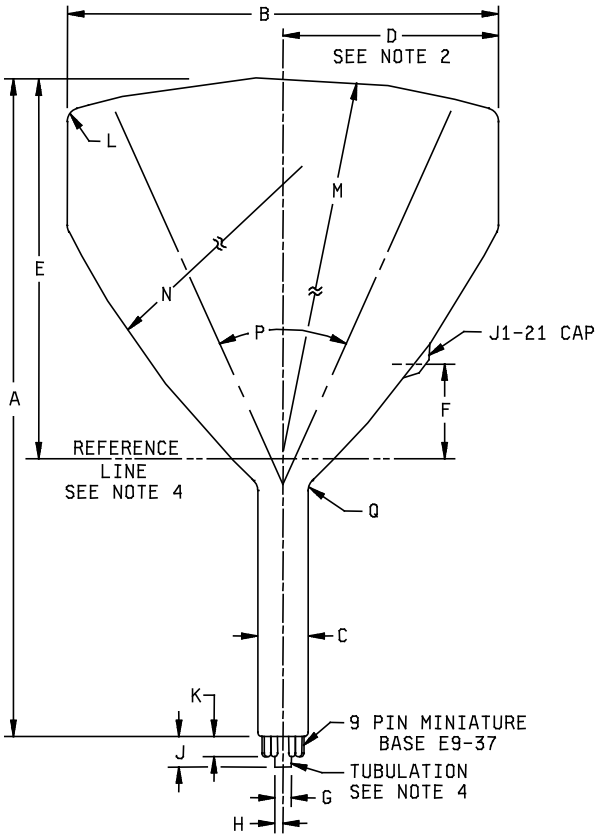
3/ The screen characteristics shall be measured with constant beam energy of 0.24 watt defocused to a spot approximately 0.25 cm in diameter. The test conditions shall be anode voltage (relative to cathode) 9,000 volts and beam current 27 μA dc maximum. The screen characteristics shall comply with the following minimum limits: cB5 = 480 cB; G5: 1 = 4.

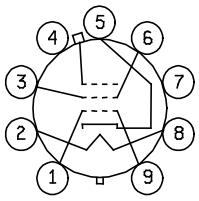
4/ The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified herein. Ec1 shall not be held at zero for more than approximately 10 seconds to prevent damage to the screen.

5/ This test is to be performed at the conclusion of the holding period.

6/ The low-pressure test shall be made with maximum voltages applied, and pressure equivalent to 20,000 feet (348 mmHg). Connections shall be made to the base pins in a manner that does not degrade the tube breakdown characteristics. Satisfactory operation shall be the absence of arc-over and harmful corona.

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BOTTOM VIEW

PIN NO.	ELEMENT
1	GRID NO.1
2	HEATER
3	GRID NO.2
5	CATHODE
6	GRID NO.2
8	HEATER
9	GRID NO.1
CAP	ACCELERATOR

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	---	16.00	---	406.40
B	10.437	10.563	265.10	268.30
C	.810	.896	20.57	22.76
D	4.500	---	114.30	---
E	9.233	9.483	234.52	240.87
F	1.915	2.415	48.64	61.34
G	---	.250	---	6.35
H	---	.125	---	3.18
J	---	.437	---	11.10
K	---	.250	---	6.35
L	.437 Rad		11.10 Rad	
M	42.000 Rad		1066.80 Rad	
N	20.000 Rad		508.00 Rad	
P	50°			
Q	.500 Rad		12.70 Rad	

NOTES:
1. Reference line is determined by the point where JEDEC G-141 reference line gauge will stop.
2. Useful screen radius.
3. Dimensions without tolerance are for information only.
4. The tubulation diameter shall be less than the diameter of the socket hole and the tubulation shall not cant so as to bear on the sides of the socket hole.

FIGURE 1. Outline drawing of electron tube type 10ANP7.

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Referenced documents: In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311.

Changes from previous issue. The margins of this specification sheet are marked with vertical line to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2020-008)

Review activities:
Navy - AS, CG, MC, OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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